Flow Marketplace — Playwright Automation Framework

UI and API test automation for the [Flow Marketplace]() web application, built with Playwright and the Page Object Model (POM) pattern.

Project Structure

```
automation/
├── pages/                              # Page Object Models
│   ├── LandingPage.js                  # /landing — marketing/home page
│   ├── LoginPage.js                    # /auth/login
│   ├── SignUpPage.js                   # /auth/signup
│   ├── ForgotPasswordPage.js           # /auth/forgot-password
│   ├── MarketplacePage.js              # /listings
│   ├── MigrationPage.js                # /listings?tab=projects (My Projects)
│   ├── MigrationDetailPage.js          # /listing/{id}
│   └── MigrationWizardPage.js          # /migration-flow (multi-step migration
wizard)
│
├── tests/
│   ├── auth/
│   │   ├── login.spec.js               # Login form + auth flow (5)
│   │   ├── signup.spec.js              # Sign up form + validation (5)
│   │   ├── signout.spec.js             # Sign out + post-logout access (4)
│   │   ├── forgotpassword.spec.js      # Forgot password page (4)
│   │   └── onboarding.spec.js          # Onboarding flow (5)
│   ├── marketplace/
│   │   ├── marketplace.spec.js         # Listing, search, card navigation (11)
│   │   ├── filters.spec.js             # Source/target filter interactions (10)
│   │   ├── pagination.spec.js          # Pagination + items-per-page control (10)
│   │   ├── explore.spec.js             # App-wide visual exploration (10)
│   │   └── list_templates.spec.js      # Inventory catalog — all templates + type (1)
│   ├── migration/
│   │   ├── migration.spec.js           # My Projects — search, filters (5)
│   │   ├── wizard.spec.js              # Migration wizard Steps 1 & 2 + detail badges
(16)
│   │   ├── validate.spec.js            # End-to-end discovery/validation walkthrough
(9)
│   │   ├── e2e_migration.spec.js       # E2E — Shopify→Lightspeed X (Customers,
linkprod01) (1)
│   │   └── e2e_migration_full.spec.js  # E2E — Shopify→Lightspeed X (3 types,
linkprod03) (1)
│   ├── api/
│   │   ├── auth.spec.js                # POST /auth/login — valid + invalid
credentials (2)
│   │   ├── marketplace.spec.js         # Listings, systems, user info (4)
│   │   ├── listing.spec.js             # Listing detail + related endpoints (11)
│   │   └── list_templates.spec.js      # API-level template catalog verification (7)
│   ├── shopify/
│   │   ├── shopifyLogin.spec.js        # Shopify store login (1)
```

```
│   │   ├── e2e_square_shopify.spec.js  # E2E — Square → Shopify (1)
│   │   └── e2e_square_shopify_v2.spec.js # E2E — Square → Shopify v2 (1)
│   ├── probes/
│   │   ├── probe_api_migration.spec.js # API migration type probe
│   │   ├── probe_lsr_wizard.spec.js    # Lightspeed Retail wizard probe
│   │   ├── probe_shopify_connect.spec.js # Shopify connect probe
│   │   └── probe_square_shopify.spec.js  # Square→Shopify probe
│   └── fixtures/
│       ├── global-setup.js             # Auth state setup (runs once before all
tests)
│       ├── customers_export.csv        # Sample Shopify customers CSV (5 records)
│       ├── products_export.csv         # Sample Shopify products CSV (5 records)
│       └── orders_export.csv           # Sample Shopify orders/sales history CSV (5
records)
│
├── utils/
│   ├── config.js                       # Central config (URLs, credentials, timeouts)
│   ├── dashboard.js                    # Custom HTML dashboard generator
│   ├── liveReporter.js                 # Live progress reporter (shows real-time test
status)
│   ├── helpers.js                      # Shared helpers: performLogin, waitForVisible
│   ├── stealth.js                      # Stealth launch args for STEALTH_MODE
│   └── testData.json                   # Static test data
│
├── test-results/
│   ├── results.json                    # Playwright JSON output (auto-generated)
│   ├── dashboard.html                  # Visual HTML dashboard (auto-generated)
│   ├── history.json                    # Run history — up to 200 entries
│   ├── inventory.json                  # Template catalog snapshot
│   └── storageState.json               # Saved auth session (auto-generated by
global-setup)
│
├── .env                                # Environment variables (do not commit)
├── dashboard.config.json               # Dashboard thresholds and history settings
├── playwright.config.js                # Playwright configuration (Chromium only)
└── README.md
```

Prerequisites

- Node.js 18+
- npm

Setup

```
cd automation
npm install
npx playwright install chromium
```

Copy `.env` and fill in your credentials:

```
cp .env .env.local   # optional — or edit .env directly
```

`.env` format:

```
BASE_URL=https://marketplace.flow.staging.linktoany.com
API_URL=https://marketplace.flow.staging.linktoany.com/api/1.0

USERNAME=your_username
PASSWORD=your_password
```

Running Tests

Command	What it runs
`npx playwright test`	All tests (Chromium)
`npx playwright test tests/auth/`	Auth tests only
`npx playwright test tests/marketplace/`	Marketplace tests only
`npx playwright test tests/migration/`	Migration tests only
`npx playwright test tests/api/`	API tests only
`npx playwright test tests/migration/wizard.spec.js`	Wizard tests only
`npx playwright test --headed`	Headed (visible browser)
`npx playwright test --grep "Login"`	Tests matching a name
`npx playwright show-report`	Open built-in HTML report
`node utils/dashboard.js`	Generate custom visual dashboard

> *Only Chromium is configured. Firefox and WebKit are commented out in `playwright.config.js`.*

Custom Dashboard

After running tests, generate the visual dashboard:

```
node utils/dashboard.js
```

This reads `test-results/results.json` and produces `test-results/dashboard.html`, which opens automatically in your browser.

Dashboard Tabs

Tab	Contents
Summary	Release gate (SAFE/REVIEW/BLOCKED/INCOMPLETE), CI context, stat cards, pass rate + sparkline, slowest tests, flaky tests

Tests	Searchable/filterable test list with expandable detail panels (errors, screenshots, traces, videos, steps, retry history)
Catalog	Migration template inventory with type filter and search
Compare	Previous vs current run diff table + full run history

Release Gate States

State	Condition
🟢 SAFE TO RELEASE	0 failures
🟡 REVIEW REQUIRED	Failures < `blockThresholdPct`
🔴 BLOCKED	Failures ≥ `blockThresholdPct` or any timeout
⚫ INCOMPLETE RUN	Any test with unknown status

Dashboard Configuration (`dashboard.config.json`)

```
{
  "slowThresholdMs": 40000,
  "blockThresholdPct": 1,
  "reviewThresholdPct": 95,
  "historyLimit": 200,
  "perTestThresholds": {
    "INVENTORY — list migration templates and type": 360000
  }
}
```

Key	Default	Description
`slowThresholdMs`	`30000`	Duration (ms) above which a test is marked slow 🦕
`blockThresholdPct`	`5`	Failure % that triggers BLOCKED gate
`reviewThresholdPct`	`95`	Pass % required for SAFE gate
`historyLimit`	`200`	Max run history entries kept in `history.json`
`perTestThresholds`	`{}`	Per-test slow threshold overrides by test title

Run History

Every dashboard generation appends a run entry to `test-results/history.json` (max `historyLimit` entries). The Compare tab uses this for trend analysis, and the Summary sparkline shows the last 10 runs.

Page Objects

LandingPage — `/landing`

Locator / Method	Description
`signInLink`	"Sign in" nav link
`getStartedButton`	"Get Started" CTA
`startFreeMigration`	Hero CTA button
`goToLogin()`	Clicks Sign In and waits for `/auth/login`

LoginPage — `/auth/login`

Locator / Method	Description
`usernameInput`	`placeholder="Enter your username"`
`passwordInput`	`placeholder="Enter your password"`
`confirmButton`	`button[type="submit"]` — "Confirm"
`forgotPasswordLink`	"Forgot password?" link
`signUpLink`	"Sign Up here" link
`goto()`	Navigate to `/auth/login`
`login(username, password)`	Fill + submit login form

SignUpPage — `/auth/signup`

Locator / Method	Description
`emailInput` / `usernameInput` / `nameInput`	Registration fields
`passwordInput` / `confirmPassInput`	Password fields
`verifyEmailButton`	"Verify email" submit button
`goto()`	Navigate to `/auth/signup`
`fillAndSubmit(data)`	Fill all fields and submit

ForgotPasswordPage — `/auth/forgot-password`

Locator / Method	Description
`emailInput`	`input[name="email"]`
`submitButton`	"Submit" button
`goto()`	Navigate to `/auth/forgot-password`
`submit(email)`	Fill email and submit

MarketplacePage — `/listings`

Locator / Method	Description
`heading`	`h1` — "Marketplace"
`templateCount`	"N available migration templates"
`searchInput`	Search input
`tabAll` / `tabMigrations` / `tabIntegrations`	Category tab filters
`previousButton` / `nextButton`	Pagination
`waitForLoaded()`	Waits for heading + first card
`getCardTitles()`	Returns `string[]` of visible card titles
`openCard(titleText)`	Clicks card by partial title
`getCardLinks()`	Returns DOM card href links for DOM inspection

MigrationDetailPage — `/listing/{id}`

Locator / Method	Description
`title`	`h1` — migration name
`migrationTypeBadge`	"File Based" or "API" badge
`getStartedButton`	Primary "Get Started" CTA
`watchDemoButton`	"Watch Demo Video"
`backButton`	"Back" nav
`waitForLoaded()`	Waits for title + badge
`clickGetStarted()`	Clicks CTA and waits for wizard
`goBack()`	Clicks Back and waits for `/listings`

MigrationWizardPage — `/migration-flow`

Handles the full 4- or 5-step migration wizard.

Generic wizard (e.g. BigCommerce/Adobe Commerce → Shopify) — 4 steps: Select → Upload|Connect → Connect → Review

Shopify → Lightspeed Retail (X-Series) — 5 steps: Select → Upload → Connect → Settings → Review

Locator / Method	Description
`step1Heading`	"Choose your data"
`step2FileHeading` / `step2ApiHeading`	Step 2 headings
`step3ConnectHeading` / `step4SettingsHeading` / `step5ReviewHeading`	Step 3–5 headings

`continueButton`	Wizard footer Continue (exact match)
`backButton`	Wizard footer Back
`customersButton` / `productsButton` / `ordersButton` / etc.	Step 1 data-type buttons
`uploadButton`	Step 2 Upload trigger
`tutorialCheckbox` / `tutorialFileInput` / `confirmFilesButton`	File upload popup flow
`storeHashInput` / `accessTokenInput` / `connectAccountBtn`	Step 2 API credentials
`connectedAccountCard` / `domainPrefixInput`	Step 3 account selection
`saveConfigButton` / `configCompleteHeading`	Step 4 settings
`startMigrationButton`	Step 5 launch
`startFreshButton` / `continueFromStepButton`	Resume modal buttons
`waitForLoaded()`	Waits for spinner to clear; auto-dismisses resume modal
`selectDataTypes(types[])`	Click one or more Step 1 data type buttons
`goToStep2()`	Click Continue to advance to Step 2
`uploadFileAndConfirm(filePath)`	Full Step 2 upload flow (single type)
`uploadFileForType(type, filePath)`	Upload scoped to a specific data type section (multi-type)
`connectDestinationAccount(domain, email, pass)`	Step 3: select existing OR full OAuth flow
`configureSettings()`	Step 4: auto-map selects, loop sub-steps, Save Configuration

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E2E Migration Tests

`migration/e2e_migration.spec.js` — Customers only (linkprod01)

Full 5-step wizard for **Shopify → Lightspeed Retail (X-Series)** with a single data type.

1. Select → Customers
2. Upload → `customers_export.csv`
3. Connect → linkprod01 (existing or new OAuth)
4. Settings → auto-configure
5. Review → Start Migration

`migration/e2e_migration_full.spec.js` — 3 data types (linkprod03)

Full migration with **Customers + Products + Sales History**.

1. Select → Customers + Products + Sales History
2. Upload → 3 CSV files

3. Connect → linkprod03 (full OAuth: store URL → login → approve)
4. Settings → 2 sub-steps (payment methods → default tax)
5. Review → Start Migration

API Reference

All authenticated requests use the header:

```
Authorization: Session {session_token}
```

Method	Endpoint	Description
`POST`	`/api/1.0/auth/login`	Login → `{ session }` (201)
`GET`	`/api/1.0/user/get-authenticated`	Current user info
`GET`	`{serviceURL}/listings`	Paginated migration listings
`GET`	`{serviceURL}/listings/{id}`	Single listing detail
`GET`	`{serviceURL}/systems`	All source/target systems
`GET`	`{serviceURL}/merchants`	Merchant info
`GET`	`{serviceURL}/feature-flags/enabled`	Feature flags

`serviceURL` = `BASE_URL/api/1.0/standalone-flow-marketplace-backend-service`

Configuration

`utils/config.js`

Key	Source	Default
`baseURL`	`BASE_URL` env	`https://marketplace.flow.staging.linktoany.com`
`apiURL`	`API_URL` env	`{baseURL}/api/1.0`
`serviceURL`	derived	`{baseURL}/api/1.0/standalone-flow-marketplace-backend-service`
`username`	`USERNAME` env	`Automation`
`password`	`PASSWORD` env	`sandeshp`

`playwright.config.js`

Setting	Value
Browser	Chromium only
Timeout (per test)	60s
Action timeout	30s

Navigation timeout	60s
Retries	1
Workers	4
Screenshots	On (all tests)
Video	On (all tests)
Trace	On first retry

Test Coverage

Suite	File	Tests
Login UI	`auth/login.spec.js`	5
Sign Up UI	`auth/signup.spec.js`	5
Sign Out UI	`auth/signout.spec.js`	4
Forgot Password UI	`auth/forgotpassword.spec.js`	4
Onboarding UI	`auth/onboarding.spec.js`	5
Marketplace UI	`marketplace/marketplace.spec.js`	11
Marketplace Filters	`marketplace/filters.spec.js`	10
Marketplace Pagination	`marketplace/pagination.spec.js`	10
Marketplace Explore	`marketplace/explore.spec.js`	10
Template Inventory	`marketplace/list_templates.spec.js`	1
My Projects UI	`migration/migration.spec.js`	5
Migration Wizard UI	`migration/wizard.spec.js`	16
Validation Walkthrough	`migration/validate.spec.js`	9
E2E Migration (single)	`migration/e2e_migration.spec.js`	1
E2E Migration (full)	`migration/e2e_migration_full.spec.js`	1
Auth API	`api/auth.spec.js`	2
Marketplace API	`api/marketplace.spec.js`	4
Listing API	`api/listing.spec.js`	11
Template List API	`api/list_templates.spec.js`	7
Shopify Login	`shopify/shopifyLogin.spec.js`	1
E2E Square→Shopify	`shopify/e2e_square_shopify.spec.js`	1

E2E Square→Shopify v2	`shopify/e2e_square_shopify_v2.spec.js`	1

Total: 124 tests across 22 files

Known Issues & Fixes

Wizard timeout cascade (fixed)

Symptom: 1 failure + 24 skips — all wizard tests skipped after one navigation timeout.

Root causes:

1. `waitUntil: 'load'` in `loginAndLoad()` blocked on all page assets loading on a slow staging server
2. `navigationTimeout` was 30s — staging server sometimes takes 30–40s
3. File-level `test.describe.configure({ mode: 'serial' })` caused one `beforeEach` failure to skip the entire file

Fix applied:

- Changed `waitUntil: 'load'` → `waitUntil: 'domcontentloaded'` (faster; content wait handled by `MarketplacePage.waitForLoaded()`)
- Added navigation retry with 3s backoff for transient network failures
- Increased `navigationTimeout` to 60s in `playwright.config.js`
- Moved `mode: 'serial'` inside each `describe` block so failures are isolated per block